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ORIGINAL ARTICLE

Early Transmission Dynamics of Spooky-like Virus 2024 (SPOK-24) in Brisbane, Australia

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ABSTRACT

BACKGROUND
In September 2024, a cluster of respiratory illness was detected in Brisbane, Queensland, associated with a novel spooky virus, provisionally named Spooky-like virus 2024 (SPOK-24). We analyzed epidemiologic data from the first hospitalized patients to estimate key parameters of transmission and guide control strategies.

METHODS
We collected clinical and epidemiologic information on the first 70 laboratory-confirmed SPOK-24 cases admitted to hospitals in Brisbane between September 26 and October 20, 2024. We estimated the incubation period, serial interval, epidemic growth rate, and basic reproductive number (R0). Doubling time was derived from the inferred growth rate using a gamma-distributed generation time with parameters estimated from known serial intervals.

RESULTS
Demographic data were limited, but no significant difference in attack rates between sexes was observed. Most early cases were associated with a long-term care facility cluster; later cases indicated community transmission. The estimated mean incubation period was 3.0 days (95% CI, 2.2 to 4.2), and the serial interval was 6.0 days (95% CI, 4.5 to 7.6). Based on a gamma-distributed generation time, the estimated exponential growth rate was 0.091/day (95% CI, 0.071 to 0.110), yielding a doubling time of 7.6 days (95% CI, 6.3 to 9.8). The basic reproductive number (R0) was estimated at 1.7 (95% CI, 1.3 to 2.3).

CONCLUSIONS
These findings confirm sustained human-to-human transmission of SPOK-24 in Brisbane. The

reproductive number and short incubation period suggest potential for continued spread without control measures. The use of generation-time-informed modeling strengthens the accuracy of early transmission metrics. Continued surveillance and non-pharmaceutical interventions remain critical to containment.